CHEMICAL ITINERARY. FROM CONTEXTUALIZATION TO GAMIFICATION: A NATURAL EVOLUTION

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Abstract

The literature abounds with examples of itineraries for humanities and science topics (botanical, geological itineraries, etc.). A connection exists between their didactic and the atmosphere in which the itinerary takes place: out of the classroom, with a reduced group of people, generating a very participative dynamic. Students often bring up questions that in other situations they might have not considered. In spite of the didactic potential, the itinerary is a tool that has not been widely exploited in chemistry. To discover the chemistry that surrounds us, some years ago we created a chemical itinerary in the city of Girona. The walk was firstly intended for primary and secondary school students, who might only associate chemistry with labs and classrooms, and also for anyone interested in learning about chemistry while visiting the city.

We present here a redesign of the itinerary to transform it into a video game. This project is a collaboration between the Faculty of Education and Psychology of the University of Girona and the Post-compulsory vocational education courses on Software Design at the Palamós High School.

Keywords: Gamification, Out-of-Classroom Learning Experience, Game-Based Learning, Chemical Itinerary, ellaboratori.cat, Learning by Walking, Mobile Learning, Location-Based Games, Augmented Reality in Education

1 INTRODUCTION

We can find some prospects in the field of education that use walks as an educational tool [1], [2]. Two types of learning are combined: learning by walking and out-of-classroom learning experience.

In this communication, we want to explain how an existing chemical itinerary has been transformed into a video game. The project involved the collaboration of students of early-years and primary teacher education at the University of Girona and computer science vocational education students at a secondary school in the nearby town of Palamós.

This itinerary, originally designed 10 years ago, has become a popular educational resource locally [3], [4]. When planning its conversion into a video game we wanted to preserve enough characteristics to make it recognizable. Furthermore, turning the itinerary into a video game takes advantage of technological resources to create a better experience. Without losing the spirit of the walk, we were able to give autonomy to the itinerary by creating a mobile video game without the need for a guide. Geolocation is used to let the users know where they are and where they should go. In this way, they can take advantage of augmented reality to solve challenges.

The project was a collaboration between the university faculty which had designed the original itinerary and a computer science student who has taken on the challenge of creating the Chemical Itinerary video game. This collaboration has provided a second learning context. The original context was intended to show that chemistry can be learned outside the classrooms, but also, the context associated with the development of this project was intended to show that video game programming could be learned outside the computer room.

1.1 Out-of-classroom learning experiences

Children's experience of enjoyment in the outdoors, particularly for young students, is widely reported [5], [6]. Informal, non-classroom-based contexts can make an important contribution to learning for pupils studying science. Whether the outcomes are measurable in terms of their contribution to the planned curriculum, or in terms of the development of the individual, their impact is significant and long lasting [7]. The importance of this teaching method is reflected by the existence of the Association of

education outside the classroom and pupils' psychosocial well-being. The Education Outside the Classroom method is gaining traction in Scandinavia, Germany, UK, United States, and other Western countries. It can broadly be defined as relocating standard curriculum teaching to places outside the buildings and walls of the schools for a single or a few days per week as a supplement to indoor classroom teaching [8]. A learning experience that could be enhanced by adding elements of a game. Encouraging, in the first step, game-based learning and then in a learning gamification experience.

1.2 Video games as an educational tool

While the influence of digital technology in formal education has been the focus of research for a couple of decades, now the ubiquitous presence of digital technology is likely to change how students engage with learning both inside and outside schools. A recent study, for instance, has shown how different technological tools shape students learning of science outside the classroom [9].

Gaming is a technological tool that has been shown to be effective as a learning tool. Literature describes two different ways of using games for education: gamification and game-based learning [10], [11]. Gamification consists of incorporating gaming elements in educational activities to improve learning, whereas game-based learning refers to learning achieved through certain educational games despite the learning not being the initial focus.

Despite the fact that game programming has become more widespread and easier to access, creating a video game aimed at learning content or skills is still relatively rare at schools and universities. One of the difficulties is to come up with the right content to be turned into a video game. In this case, this was already in place. The itinerary has proven an effective and motivating learning experience for students of different school levels in Girona for the past 10 years. Turning it into a video game was a natural challenge.

1.3 Gamification

In 2002 Nick Pelling, a British game developer, coined the term gamification. He defined this term as the interest in applying game concepts in the user interfaces of the applications he created to make electronic transactions more enjoyable and faster [12]. It is not until after 2010, that Kevin Werbach refers to gamification as "the use of game elements and game design techniques in non-game content" [13].

The conversion of the itinerary into a video game comes from a collaboration project between the University of Girona (students of Education) and the high school in Palamós (students of Computer Science). This articulated collaboration represented a gamified learning experience among students.

2 METHODOLOGY

2.1 Learning by Walking

In the literature, we find many examples of itineraries to develop different fields of knowledge. An excellent example of combining Learning by Walking and video games comes from Mid Sweden University [1].

2.2 Articulating the collaboration as a gamified learning experience

Teachers in training developed this project within the subject: 'Video Games and Education' from the fourth year at the Faculty of Education and Psychology of the University of Girona, while the computer science students are in their final year and undertake this project in module 15, Dual. The project was scheduled during the first four months (from September to January).

The design of the video game based on the Chemical Itinerary of Girona was one of several videogames designed through the collaboration between students of Early-ages and Primary Education and Computer Science vocational secondary school. These two groups of students collaborated, mainly through a website designed by one of the authors, called http://ellaboratori.cat. Students used the website to communicate and to follow their respective courses. The website is currently divided into two parts: the game and the public website. The public website is open to everyone and all the designed games can be downloaded. Those taking part in the courses can only access the game part and this developed as the course progressed.

This means that the learning experience was also gamified. The game was based on a gamification model used by one of the authors[1]. This model was previously used in the subject "Video Games and Education". The model present on the website has been used in three different phases. In the first phase, it was used as a way of introducing the players (students) to each other and to discuss their collaborative projects. This involved introductory videos, tutorials and examples of games from previous years. In the second phase, it was used to send video game proposals from Teaching students to Computer Science students. In the third phase, video game prototypes were made available to in-service teachers, thus seeing their development.

2.3 The Chemical Itinerary video game

Most of the games developed were intended to provide teachers with an educational classroom resource. They create a classroom experience, which can help the teacher to develop new teaching strategies based on the game. The case of the Chemical Itinerary was slightly different: it was intended to be used outside of the classroom. Furthermore, one of the authors participated in this project as a continuation of some work that started few years ago.

The fact that the idea for the video game was based on a tried and tested resource meant that the narrative was much clearer, and the educational goals were better defined than most of the other collaborations. Unfortunately, when the project was in its final phase, the Covid-19 pandemic stopped the tests that were being carried out in the field and left the project in the prototype stage. A prototype that can be downloaded, tested and played, but still pending the integration of a final product.

2.4 Adapting a PBL methodology to the design of Video Games

This project is based on Project Based Learning (PBL). The project to be carried out is the creation of a video game for high school students in vocational training in Computer Science based on the design from students in early childhood and primary education. A project that lasts one academic year.

If we delve deeper into the project we are developing, we could define this learning as 'Creation of Video Games-Based Learning'. A type of learning that has many elements in common with Game-Based learning (GBL). However, in this project we are more interested in the creation of the game than the game itself. For the creation of our video games, we use narratives, dynamics, mechanics and aesthetics of existing games.

A simple way to define the concept of Creation of Video Games-Based-Based Learning would be halfway between Game-Based Learning and Gamification. In carrying out this project, our students are, effectively, learning to gamify. A shared learning with teachers.

As the product of this collaboration, we obtained games that are meant to be played in primary and early years classrooms. Depending on the relationship between the content we want to develop and the game we want to design, we could differentiate three types of video games in our creations:

- GBL (Game-Based Learning), where the content of the subject is directly associated with the game.
- sGBL (Serious Game-Based Learning), where the game is the content of the subject to be developed.
- KBL (Kahoot-Based Learning). These video games do not achieve either of the above two goals and will never be taken to any classroom. However, they are very useful in this project because, as in video games, when you lose, you come back and try harder.

We could include the Chemical Itinerary video game as Serious Game-Based Learning because the contents that are discovered during the game are the game itself.

2.5 Mobile Learning

There has been a lot of literature written about Mobile Learning in the last twenty years. Learning is mediated by knowledge and technology as instruments for productive enquiry, in a mutually supportive and dynamically changing relationship [14].

[1] This gamified module (The Intergalactic Conspiracy) is accessible at: https://educaciodema.cat/recurs/conspiracio-intergalactica-gamificant-educacio-superior-pere-cornella

2.6 Location-based games

The most important point when turning the itinerary into a video game was that it could offer us the technology that the original itinerary did not have. In addition, geolocation is probably the most important aspect as the itinerary ceases to be a guided tour and becomes autonomous. The video game allows you to do the itinerary without a guide. It also has two operating options, for a group of students or individually.

2.7 Augmented Reality in Education

Another aspect to highlight is augmented reality. Integrating elements of the video game into the reality around you when taking this *walk through the invisible* enhances the experience. The user will visualize the environment around him on screen, along with the main character of the video game and use augmented reality to solve some of the challenges and puzzles that will be encountered along the way.

3 Results

3.1 The Chemical Itinerary video game

The design of this video game can be seen as a process of transformation of the existing itinerary. At some point, the original idea of the itinerary had to be changed, but the educational narrative structure was mostly unaltered. Each of the transformed and added elements went through a test period to assess whether it was worth incorporating them into the video game or whether it was better not to use them.

The first completed and functional version (http://ellaboratori.cat/labflix/) presented the following features:

· The story remains the same. The information for each stop is the same as that offered in the itinerary. As are the riddles and challenges, we encounter at each of the stops. This allows us to meet the aim of designing a game that will be familiar to the schools that have been using the itinerary these past 10 years.

· Geolocated Game. The 11 stops of the itinerary are geolocated and marked on the map of Girona. The game recognizes the point where the player is at any given time and indicates the next point on the route.

· Augmented Reality. The player of our itinerary appears on the map following our steps. The player will use Augmented Reality to resolve challenges that they find on the walk.

· Different ways to play. The game allows the player to start the game directly without any registration or login, and it allows teachers to create a group with their students and become a competitive or a cooperative game.

3.2 The collaboration as a gamified learning experience

One of the purposes of this article is to describe the conversion of the Chemical Itinerary to a video game. A video game that was born from a project between two different educational stages and that has given rise to other video games and game proposals. In the two years of collaboration, more than thirty video games have been created. From an interactive digital whiteboard, to an anthill where the teacher is the queen ant and creates tasks for the group, or a maths escape room, or inside a chemistry lab. We describe three of these video games to let the reader know what kind of products this collaboration generates.

3.2.1 Botanic GO (Prototype complete)

Using the same dynamics as *Pokemon GO*, a botany game has been developed from ellaboratori.cat for any educational stage. The teacher proposes weekly challenges and students have to find the species in the natural world. They send a photograph, and the teacher validates it. A proposal that can go from taking a photo of a leaf or a flower in early childhood education, to creating a virtual herbarium at university.

3.2.2 Among US in a zoo (under construction)

The possibilities that a game like Among Us gives us are incredible. In this case, it is a hunter who wants to catch the animals in a zoo. The strength of this game is based on communication between players. It has to be fast, concise and direct. Taking advantage of this strong point of the game, we want to implement it in two classes that speak different languages.

3.2.3 Creating your own world (Special virtual reality project for next year)

Virtual reality has probably landed in classrooms because of its potential. This project, which has already begun to be devised, will be developed in special education classrooms. Moreover, our students will be able to create their world by drawing it with their fingers in space, sharing it with other students and living other worlds.

4 CONCLUSIONS

The project has demonstrated the potential for gamified collaboration between university and high school students. In particular, the strengths of the web-based gamified modules that were shared as a collaboration platform by students from both groups. This collaboration resulted in different video games that, according to the relationship between the content and the game, the author groups into three possible categories: GBL, sGBL, KBL.

As part of this experience, one of the authors proposed as one of the video-game projects, the adaptation of an existing chemical itinerary in the city of Girona. This served as a case study on how the PBL-based initiative with students of education could also be used with in-service teachers.

The three strongest points of the Chemical Itinerary, the video game, have also been detailed. Mobile Learning, Location-based games and Augmented Reality in education.

The final product has shown how educational resources can be adapted and improved by turning them into video games, as well as, compared to other video games produced, providing information about the importance of educational narrative as a basis for educational video games.

4.1 Future development of the video game

One of the strengths of this project is that the video game survives the students who created it and the Itinerary continues to grow.

We are working on a second phase of the project where we will incorporate the virtual figure of Prof. Duran, the creator of the Chemical Itinerary. This character will be the one who will tell, at each of the stops along the route, the story hidden at each point.

Another line of development of the itinerary is that the participants will obtain points when they solve the enigmas posed. We will also add a counter that indicates the time that has been designated to solve the enigmas.

We are also working on the addition of hidden places within the itinerary where small chemical experiments can be done, and the results can be compared with those of a virtual chemical laboratory that will also be in the game.

The use of augmented reality will allow, for example, the virtual visit of *Cromats Ensesa* (a chrome-plate workshop which no longer exists) or to see the Cathedral's angel up close.

In addition, how far can the process of transformation go? The creation of the Chemical Itinerary video game from the real chemical itinerary is fantastic because it is like learning: a process with successes and failures, which takes place in a real environment, and where the most important thing, as in the itinerary, is the journey that is being taken.

4.2 A stronger collaboration over time

The collaboration between the training of early-years and primary school teachers and vocational secondary education in Computer Science has become more powerful as time goes on.

4.2.1 ellaboratori.cat, a new website every year

The website was created in the first year as a communication tool between the two groups of students. On one hand, to teach what types of video games could be made from the Palamós high school and on the other, what ideas came from the University of Girona. The second year it also allowed the curricular proposals to be classified and became the evaluation tool and the testing interface for video games as they were developed. In the next update, the website will be another element in the process of gamification of the two groups.

4.2.2 One gamified learning experience, two more united narratives

As described in section 2, this collaboration is based on a gamified experience, the Intergalactic Conspiracy of the University of Girona. Currently, a second narrative is being integrated into the Conspiracy, the Dark Men from the high school in Palamós.

4.2.3 The biggest goal

This article has two objectives: to describe the conversion of the Chemical Itinerary into a video game and to describe the gamified collaboration from which this video game was born. The main goal of this collaboration is the learning by students who design and create video games rather than the product itself, video games. On the one hand, computer science students have the opportunity to develop a product with a client and teachers in training to develop a methodology based on video games in their future classrooms.

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